MILITARY STANDARD

MIL-STD-1386-10
(Navy)

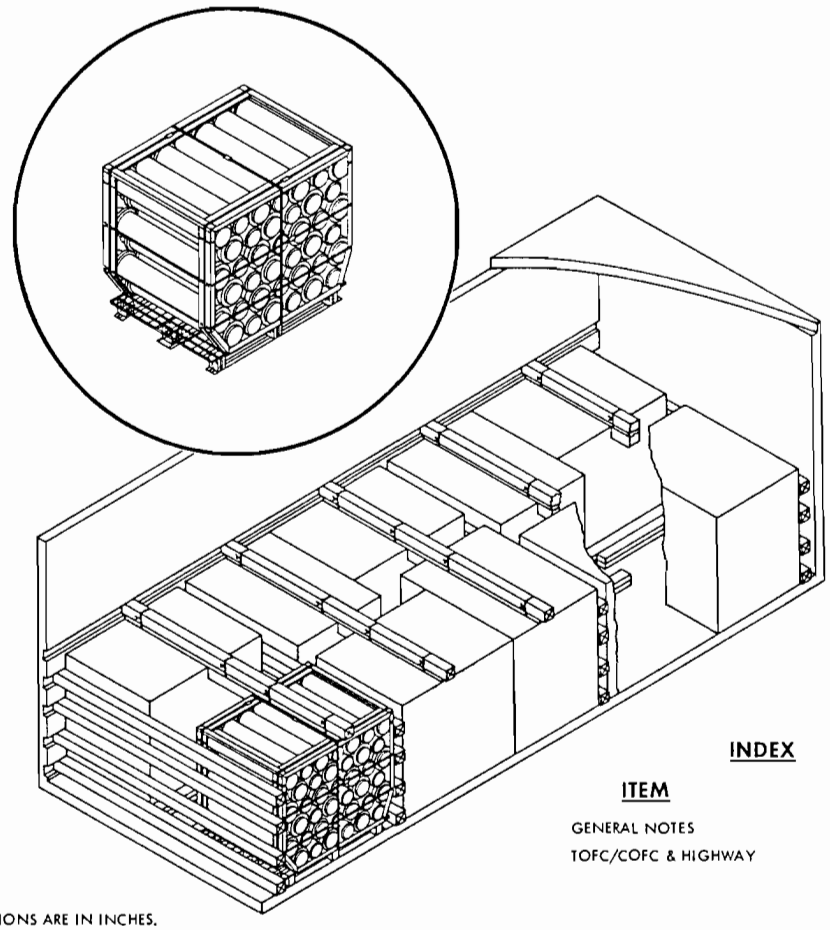
24 NOVEMBER 1972

LOADING OF HAZARDOUS MATERIALS AND ASSOCIATED ITEMS IN MILVAN CONTAINERS

5"/54 CARTRIDGE
FLEET ISSUE UNIT LOAD

UNIT LOAD DATA

UNIT LOAD DRAWING WR-54/5
WEIGHT . . . . . SEE TABLE I
DIMENSIONS. . . . . 48 L x 36 1/2 W x 43 3/8 H
CUBE . . . . . 43.9 CU FT
DOT HAZARD CLASSIFICATION. . . . . EXPLOSIVES B
COAST GUARD CLASS . . . . . II-B



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TOFC/COFC & HIGHWAY	2 & 3

NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX OF STANDARDS, MIL-HDBK-236.

AUTHORIZED FOR TRAILER-ON-FLATCAR (TOFC), CONTAINER-ON-FLATCAR (COFC), CONTAINERSHIP AND HIGHWAY SHIPMENTS

M.S. Gray NWHL 10/19/72
SIGNATURE TECHNICAL DIRECTION AGENT(TDA) DATE

J E Kelly 10/24/72
SIGNATURE ORD SYSCOM, BY DIRECTION DATE

APPROVED BY US COAST GUARD

DATE 9 AUGUST 1974

APPROVED BY BUREAU OF EXPLOSIVES

A. F. Grassmuck
DATE 9/28/72

ORIGINATOR L R Caringi
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

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GENERAL NOTES

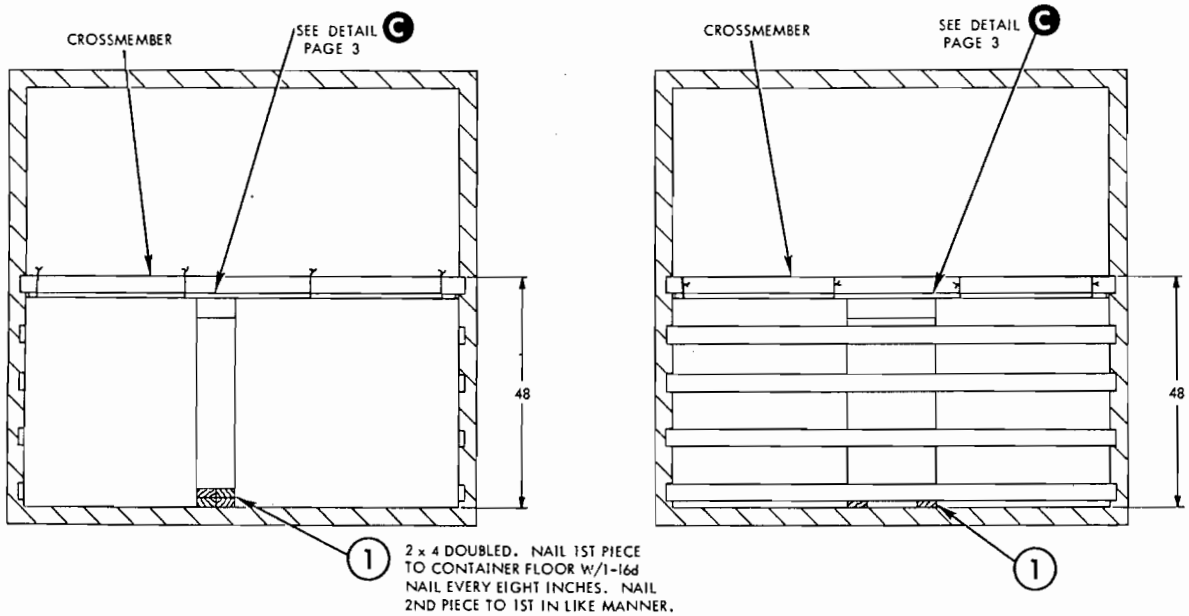
1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR LOADING THE MILVAN CONTAINER WITH FLEET ISSUE UNIT LOADS OF 5"/54 CARTRIDGES IN MK 14, MK 14 MOD 1, MK 21 MOD 0 AND REDUCED MK 14 TANKS. THE UNIT LOAD SHALL BE IN ACCORDANCE WITH WR-54/5.
2. THE LOADED CONTAINER MAY BE TRANSPORTED OVER PUBLIC HIGHWAYS ON THE MILVAN CHASSIS USING THE FOLLOWING CONFIGURATION:
   A. CONTAINER ON 20-FT MILVAN CHASSIS WITH A DOUBLE BOGIE LOCATED IN THE NEXT TO LAST HOLE AND PULLED BY A TRACTOR WITH A DOUBLE DRIVE AXLE.
3. THE LOADED CONTAINER MAY ALSO BE TRANSPORTED OVER PUBLIC HIGHWAYS USING COMMERCIAL EQUIPMENT OF APPROPRIATE CAPACITY.
4. LOADED MILVAN CONTAINERS MUST BE ON 20-FT MILVAN CHASSIS EQUIPPED WITH TWO BOGIE ASSEMBLIES WHEN BEING MOVED IN TOFC SERVICE.
5. THE MILVAN SHALL BE EQUIPPED WITH A MECHANICAL LOAD-BRACING SYSTEM THAT MEETS THE REQUIREMENTS OF THE BUREAU OF EXPLOSIVES PAMPHLET 6C AND SHIPPED ON FLAT CARS THAT MEET THE REQUIREMENTS OF PAMPHLET 6C.
6. IN ADDITION TO THE BELT RAILS REQUIRED BY PAMPHLET 6C, THE MILVAN CONTAINER HAS BELT RAILS AT THE 72" AND 83" HEIGHTS FROM THE CONTAINER FLOOR.
7. CONTAINERS ARE EQUIPPED WITH 25 CROSSMEMBERS. UNUSED CROSSMEMBERS MUST REMAIN WITH THE CONTAINER AND SHALL BE FASTENED INTO BELT RAILS FOR SHIPMENT.

CAUTION

WHEN POSITIONING CROSSMEMBERS, MAKE A VISUAL CHECK TO ENSURE THAT CROSSMEMBERS ARE LOCKED SECURELY IN PLACE.

8. THE DUNNAGE SPECIFIED IS OF NOMINAL SIZE. FOR EXAMPLE, 1 x 4 MATERIAL IS ACTUALLY 3/4 THICK BY 3 5/8 WIDE AND 2 x 6 MATERIAL IS ACTUALLY 1 5/8 THICK BY 5 1/2 WIDE.
9. WHEN THE VOID BETWEEN THE CROSSMEMBER AND THE UNIT LOAD IS GREATER THAN ONE INCH USE FILL MATERIAL TO SHIM BETWEEN CROSSMEMBER AND LOAD AS SHOWN IN DETAIL D.
10. BRACE, PIECE 3, SHALL FILL THE VOID BETWEEN THE UNIT LOADS.
11. APPLICABLE MATERIAL SPECIFICATIONS:
    DUNNAGE LUMBER - FED. SPEC. MM-L-751
    NAILS-FED. SPEC. FF-N-105, COMMON CEMENT COATED
12. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CONTAINER IN AN ACCESSIBLE AREA, CLOSE AND SEAL CONTAINER DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF THE CONTAINER FOR THE APPROPRIATE MODE OF TRANSPORTATION AS PRESCRIBED IN OP 2165 (VOL. 1).

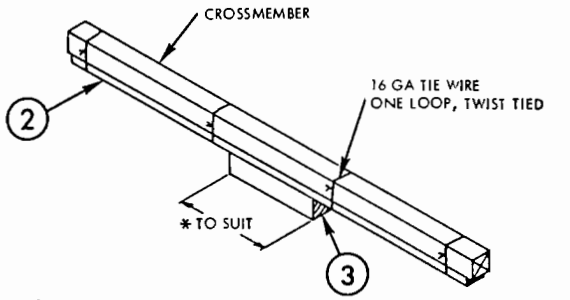
COFC, TOFC, CONTAINERSHIP & HIGHWAY



SECTION A-A

SECTION B-B

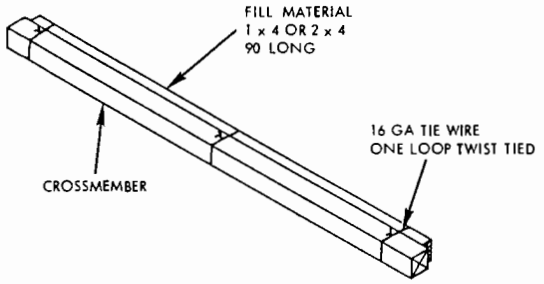
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* SEE NOTE 10
PAGE 2

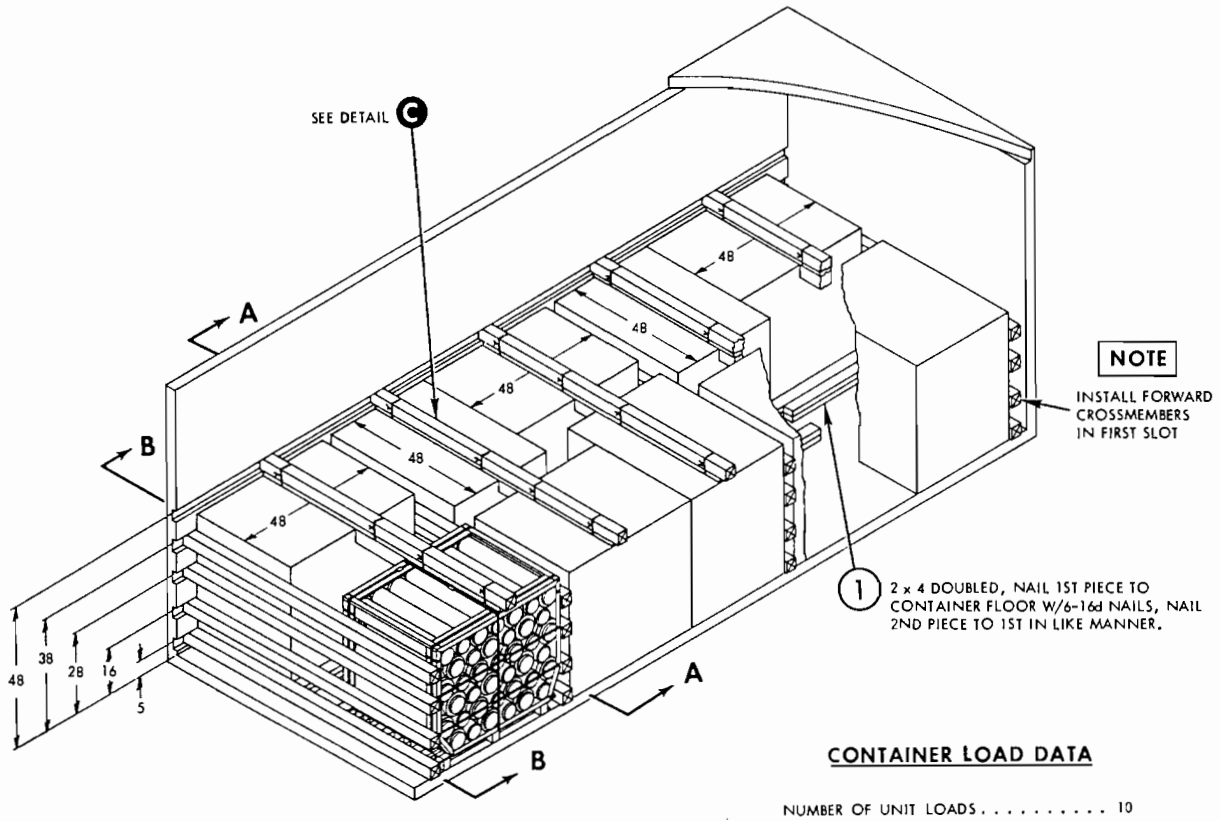
DETAIL C

HOLD DOWN
5 REQ'D



DETAIL D

FILL MATERIAL
SEE NOTE 9
PAGE 2



CONTAINER LOAD DATA

NUMBER OF UNIT LOADS 10
NUMBER OF CROSSMEMBERS REQ'D 21
LOAD WEIGHT . . . . . SEE TABLE I
DUNNAGE WEIGHT . . . . . 233 LBS
CONTAINER WEIGHT . . . . . 5700 LBS
GROSS WEIGHT . . . . . SEE TABLE I

TABLE I

TANK	UNIT LOAD WT (LBS)	CONTAINER LOAD WT (LBS)	GROSS WT (LBS)
MK 14	1676	16,760	22,693
MK 14 MOD 1	1770	17,700	23,633
MK 21 MOD 0	1789	17,890	23,823
REDUCED MK 14	1399	13,990	19,923

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
3	SWAY BRACE	4 x 4 TO SUIT	5	SEE 2	–	–
2	SHIM	4 WIDE x 90 THICKNESS TO SUIT	5	3	3	16d
1	SLEEPER	2 x 4 x 48	20	SEE SECTION A-A		

LIST OF MATERIALS & NAILING DATA

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